

PseudoRandomness

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Reading notes from Justin Thaler's "Proofs Argument and Zero Knowledge"

Keywords:

1 What is PseudoRandomness?

- Perfect secrecy $\xrightarrow{\text{Computational, lax}}$ INDISTINGUISHABILITY
- True random $\xrightarrow{\text{Computational, lax}}$ Pseudorandomness
- Refers to a distribution of strings. Distribution $\mathcal{D}$ over strings of length l is pseudo random means that $\mathcal{D}$ is IND from uniform distribution over strings of length l .
- And any polynomial time algorithm can **NOT** differentiate between a string from uniform distribution and string from pseudorandom

What is true randomness? Is it related to unpredictability, let's define that. ANS. True randomness is a sequence of bits that cannot be predicted or reproduced, even with complete knowledge of the algorithm used to generate it. It is often derived from physical processes, such as thermal noise or radioactive decay, which are inherently unpredictable.

2 Pseudo Random Function

3 Pseudorandom generator

4 Pseudorandom permutation